GLOBAL TECHNOLOGY COMPETITION

Official Rulebook

A strategy game of science, talent and secrets for four teams

Players	Rounds	Round 1	Later rounds
4 teams + 1 game master	Set by the GM (not announced)	20 minutes	10 minutes each

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CHAPTER 1

The Goal

You lead the science and technology program of a nation. Get the highest score at the end of the game to win.

Four teams play: the United States, China, France and Russia. A game master (GM) runs the game. Each round, your team spends a budget on education, research, spying and immigration policy.

Your score comes from three things:

- **Technology Points (TP).** You earn them from research. Harder research gives more points.
- **Tacit Knowledge (TK).** This is the know-how you get from doing research. High TK multiplies your TP.
- **Wealth (W).** More wealth at the end adds a small bonus.

Most of your choices are secret. Basic research and caught spies are public.

Each team member holds a portfolio, such as Education or National Security. **Appendix C** shows which numbers each portfolio should watch.

> **IMPORTANT**
> The GM does not announce the last round. Plan for a game that can end at any time.

CHAPTER 2

Key Terms

Learn these twelve terms first. The rest of the rulebook uses them.

Term	Meaning
W	National wealth. Your budget each round is 2 × W.
Dice	Your spending points. One die = one point.
K	Talent pool (human capital). Education raises it.
SE	Secondary education. Buying it raises K.
TE	Tertiary (university) education. Buying it raises K.
TK	Tacit knowledge. Research raises it. High TK makes research pay more and multiplies your score.
BR	Basic research. Public. It raises TK and unlocks harder research.
BR effective	Your own BR total + 0.75 × the BR totals of all other teams.
AR	Applied research, at level L, M or H. Secret. This is how you earn TP.
TP	Technology points. The main part of your score.
SP	Espionage. You spy on a rival to see or steal its TP.
IM	Immigration policy. Open (1) or closed (0).

Each country also has a **regime**: "demo" (democracy) or "auto" (autocracy). It changes some dice results a little.

CHAPTER 3

The Four Countries

The countries are different on purpose. No country is best. Each has a real path to victory.

Starting values

Country	W	Dice per round	Start K	Start TK	IM	Regime	SE cost	Talent flow at start
US	6	12	6	0	Open	demo	1	1.79
China	5	10	7	0	Closed	auto	3	1.02
France	4	8	6	4	Open	demo	1	0.60
Russia	4	8	5	7	Closed	auto	1	0.60

"SE cost" is the price in dice to buy secondary education once. "Talent flow" is the number that multiplies your K each round. See Chapter 8.

Strengths and weaknesses

Country	Strengths	Weaknesses
US	Largest budget. Strong universities pull in talent.	Open borders make it easy to rob.
China	Large population. Secondary education gives the most talent.	Secondary education costs 3 dice.
France	Starts with 4 TK. Education is cheap.	Small budget. Open borders make it easy to rob. Talent drains away.
Russia	Starts with 7 TK. Closed borders stop theft.	Smallest budget. Talent drains away.

CHAPTER 4

How a Round Works

All teams play at the same time. You do not see what rivals choose until the round ends.

Time limits

Part	Time
Round 1	20 minutes
Each later round	10 minutes
GM resolves the round; you read your results	About 3 minutes

Steps in each round

1. Discuss your plan. Decide with your team's voting rule.
2. Enter your spending on the team screen. The budget counter shows the dice you have left.
3. Write one or two sentences on why you chose this plan. You cannot submit without them.
4. Click **Submit** before the timer ends. Check your choices before you click.
5. The GM resolves the round. The game does these steps, in this order:
    a. Immigration changes and education
    b. Talent flow
    c. Basic research
    d. Applied research
    e. Espionage
6. Read your private log. It shows your real results.
7. The GM starts the next round. Saved budget now adds to your W.

> **TWO RULES TO REMEMBER**
>
> **The dice table on your screen is only a preview.** The GM rolls the real dice. Your private log shows the real results.
>
> **If you do not submit in time, your team does nothing that round.** You also save nothing.

CHAPTER 5

Spending Options

Your budget each round is 2 × W dice. You can spend all of it, part of it, or none of it.

Option	Cost (dice)	Effect
SE (secondary education)	1 (China: 3)	K goes up by 1.02 (US), 1.37 (China), 0.74 (France) or 0.87 (Russia). Once per round.
TE (tertiary education)	1	K goes up by 1. If you are open, K goes up by 2. Once per round.
IM change	0	You open or close immigration.
BR	1 per die	See Chapter 6. No limit.
AR	Rises with each die (see below)	See Chapter 6.
SP	1 per target	See Chapter 7. Maximum 2 targets per round.

Applied research costs

Each extra AR die you buy in a round costs more than the one before. This rule makes a big push in one round expensive.

Three things to know:

- **One count for all levels.** L, M and H dice add to the same count.
- **The count starts again each round.** In a new round, your first AR die costs 1 again.
- **There is a limit.** You can buy at most 3 L dice, 2 M dice and 1 H die per round. That is 6 AR dice in total.

AR dice you buy this round	1	2	3	4	5	6
Total cost (dice)	1	2	4	6	9	12

The first 2 AR dice cost 1 each. The next 2 cost 2 each. The last 2 cost 3 each.

You buy this round	AR dice	Total cost
1 M die	1	1
2 L dice	2	1 + 1 = 2
2 L dice + 1 M die	3	1 + 1 + 2 = 4
3 L + 2 M + 1 H dice	6	1 + 1 + 2 + 2 + 3 + 3 = 12

TIMING CHANGES THE PRICE

To buy 4 AR dice in one round costs 6. To buy 2 AR dice in each of two rounds costs 2 + 2 = 4.

To spread AR across rounds costs less. To buy a lot in one round costs more, but you can get results sooner.

CHAPTER 6

Research

Basic research unlocks harder work. Applied research earns TP.

Basic research (public)

- Each BR die succeeds on a roll of 4, 5 or 6.
- Each success adds 1 to your BR total and 1 to your TK.
- **Talent bonus:** if you buy at least 1 BR die, you get extra successes. The number is K ÷ 10, rounded down.
- **Regime:** a demo country multiplies its successes by 1.1. An auto country multiplies them by 0.9. The result is rounded.
- All teams see your BR successes and your rolls.

You also gain from the BR of other teams. Your **BR effective** = your own BR total + 0.75 × the BR totals of all other teams. You need BR effective to unlock M and H research.

Applied research (secret)

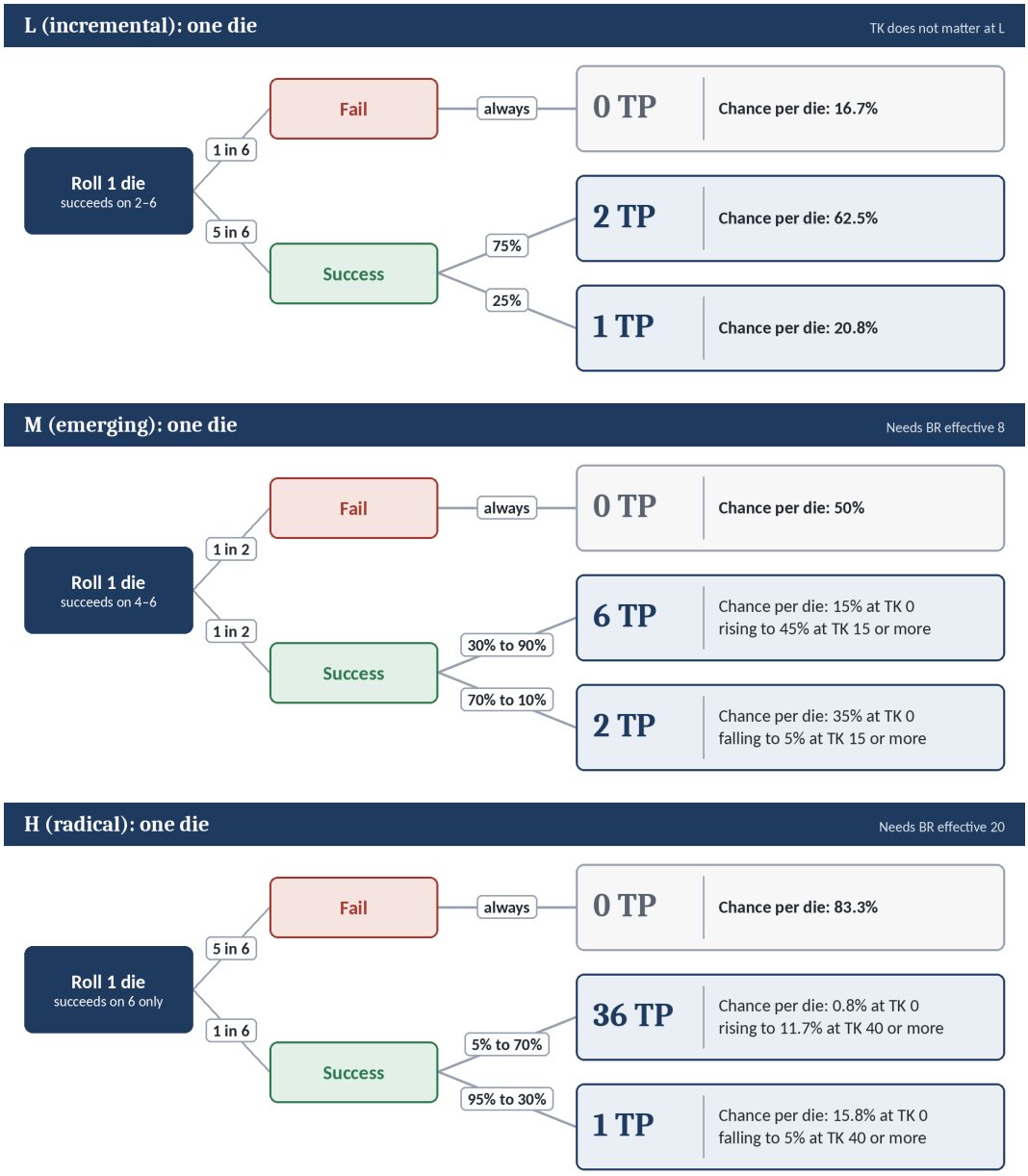
Level	BR effective needed	A die succeeds on	TP for each success	Max dice	Locks at	Weight
L (incremental)	0	2–6 (5 in 6)	2 (75% chance), otherwise 1	3	10 TP	1
M (emerging)	8	4–6 (1 in 2)	6 or 2. The chance of 6 goes from 30% (TK 0) to 90% (TK 15).	2	20 TP	2
H (radical)	20	6 (1 in 6)	36 or 1. The chance of 36 goes from 5% (TK 0) to 70% (TK 40).	1	40 TP	3

More applied research rules

- **Every AR die adds 1 TK, even if it fails.**
- **Locks:** when your TP at a level reaches the "locks at" number, you cannot buy more AR at that level.
- **Talent limits:** for M research, if K is below 7, your TK counts as 10 lower. For H research, if K is below 12 or TK is below 20, your TK counts as 10 lower.
- **Regime at M and H:** an auto country multiplies its successes by 1.1. A demo country multiplies them by 0.9. The result is rounded.
- Your screen opens the M and H boxes only when your BR effective is high enough.

What one AR die can do

Each AR die has three possible results: it fails, it gives the small reward, or it gives the big reward. The trees show the chance of each result. At M and H, high TK makes the big reward more likely. The talent limits can lower the averages in the table.



Level	Average TP per die, TK 0	Average TP per die, high TK
L	1.46	1.46 (TK does not matter)
M	1.6	2.8 (TK 15 or more)
H	0.46	4.25 (TK 40 or more)

CHAPTER 7

Espionage

You can steal points from an open country, but not a closed country.

The key advantage of espionage is being able to see what AR categories other countries have been investing in, and how much. This will allow you to deduce their strategy and estimate their TP score.

- Really lucky rolls have some chance of copying what they have learned in that category, adding their AR to your own. Really unlucky rolls will allow other countries to catch and identify you.

For each spy, choose a target country and a level (L, M or H). The game rolls one die for each spy. It adds 1 if the target is open. It subtracts 1 if the target is closed. The result stays between 1 and 6.

Results

Adjusted roll	Spy on L	Spy on M	Spy on H
Caught	1	1–2	1–3
Fail	2–4	3–4	4
Observe	5	5	5
Steal	6	6	6

- **Caught:** all teams see a public message about your spy.
- **Fail:** nothing happens. Only you know.
- **Observe:** you see the target's TP at that level.
- **Steal:** you see the target's TP at that level. Then your TP at that level becomes the target's TP from the start of this round, if theirs is higher. The two numbers do not add.

Chances

Chance per spy	Open target	Closed target
Steal	1 in 3	Never
Observe	1 in 6	1 in 6
Caught, spy on L	Never	1 in 3
Caught, spy on M	1 in 6	1 in 2
Caught, spy on H	1 in 3	2 in 3

CHAPTER 8

Talent Flow and Immigration

Talent moves toward rich countries with strong universities.

Each round, the game multiplies your K by your talent flow number. The effect builds up: a number above 1 grows your K every round, and a number below 1 shrinks it every round.

Your talent flow number depends on three things:

- Your W × your university strength, compared with the other teams.
- Your immigration policy. A closed country cannot pull in talent from other countries, so its number stays at or below 1.
- The choices of the other teams.

Talent flow numbers

The first row shows each country's talent flow number **at the start.** The other rows show how much each number changes if one country changes its immigration policy and the others do not. Values can also change based on other policies (e.g. investing in university infrastructure will net more talent flow).

Situation	US	China	France	Russia
Start (number)	1.79	1.02	0.60	0.60
Change if US closes	−0.53	+0.24	+0.14	+0.14
Change if China opens	−0.04	+0.07	−0.02	−0.02
Change if France closes	0	0	0	0
Change if Russia opens	0	0	0	0

Example: if the US closes, its number falls from 1.79 to 1.26. China's number rises from 1.02 to 1.26.

For France and Russia, a policy change does not change talent flow. It changes only the theft risk and the TE bonus. (This could change as France or Russia begin to invest in stronger universities!)

When an immigration change takes effect

- It protects you from theft (or exposes you) in the same round.
- It changes your talent flow from the next round.
- It changes the effect of TE at once (+1 K if closed, +2 K if open).

Talent flow also changes when any team's W changes.

CHAPTER 9

Savings & Interest Accumulation

Dice you do not spend make your country richer for the rest of the game.

At the start of the next round, 2/3 of your unspent dice adds to your W. The game keeps fractions and adds them later. Your budget is 2 × W, so a saved die gives more dice in every later round.

Dice left unspent	W gain	Extra dice in every later round
3	+2	+4
6	+4	+8
12	+8	+16

RISK

Dice you save in the last round are lost. The GM does not announce the last round. Each time you save, you take that risk.

CHAPTER 10

Scoring and the End of the Game

The team with the highest score after the last round wins. The GM announces the final scores.

The score formula

$$\text{Score} = (\text{L TP} + 2 \times \text{M TP} + 3 \times \text{H TP}) \times \text{TK multiplier} + 2 \times \sqrt{\text{W}}$$

The TK multiplier is 1 + 0.2 × (natural log of (1 + TK)). It grows fast at first, then slowly.

TK	0	10	100	1,000
TK multiplier	1.00	1.48	1.92	2.38

Worked example

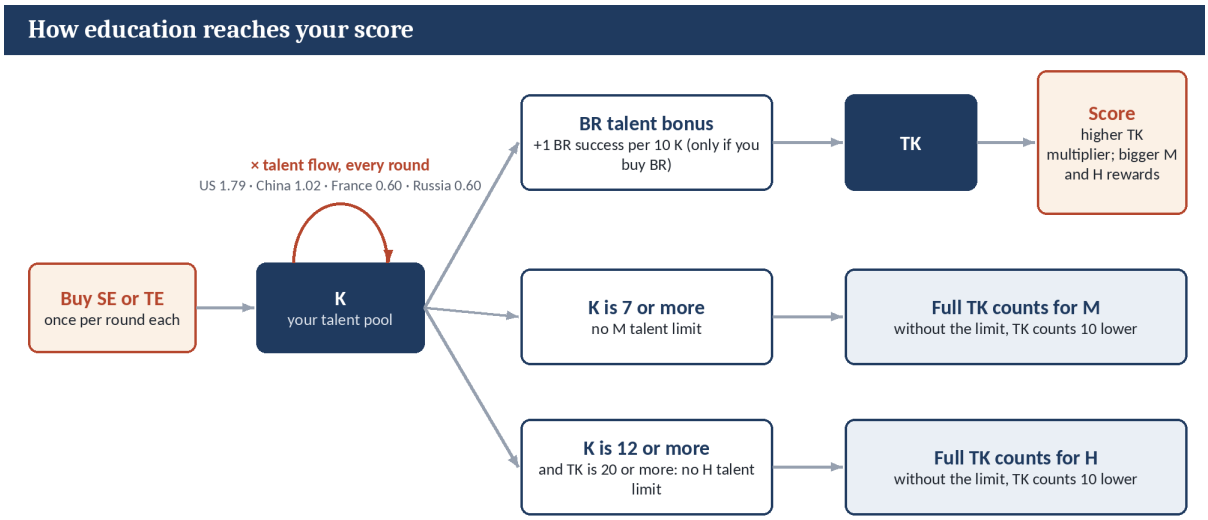
Step	Value
TP at L, M, H	10, 20, 0
Weighted TP	10 + 2 × 20 + 3 × 0 = 50
TK	30 (multiplier 1.69)
TP part	50 × 1.69 = 84.3
W	6 (wealth part 2 × √6 = 4.9)
Score	84.3 + 4.9 = 89.2

Your screen shows your own TP, K, TK and W. It does not show your score or the TP of rivals. Use spies to learn what rivals have.

CHAPTER 11

Education and Talent

Education builds talent (K). Talent reaches your score in three ways. Talent flow can grow it or drain it every round.



Education never adds TP directly. It works only through K.

What one purchase buys

Purchase	Cost (dice)	K gained
SE (secondary)	1 (China: 3)	US 1.02, China 1.37, France 0.74, Russia 0.87
TE (tertiary)	1	1 if your borders are closed, 2 if open

What K does

Use of K	Effect
BR talent bonus	+1 BR success per round for each 10 K. Only if you buy at least 1 BR die.
M talent limit	If K is below 7, your TK counts 10 lower for M research.
H talent limit	If K is below 12 (or TK is below 20), your TK counts 10 lower for H research.

Talent flow changes the value of a school

Talent flow multiplies all of your K every round, including the K you just bought. So the same purchase grows in one country and drains away in another. The table shows what remains of the K from one TE purchase.

Country	K at purchase	After 1 round	After 2 rounds	After 4 rounds
US (open, × 1.79)	2	3.6	6.4	20.5
China (closed, × 1.02)	1	1.0	1.0	1.1
France (open, × 0.60)	2	1.2	0.7	0.3
Russia (closed, × 0.60)	1	0.6	0.4	0.1

These numbers use the talent flow at the start. They change when any team changes its W or its immigration policy.

Example: the US and France each buy TE once. After 4 rounds, the US purchase has grown to about 20 K. That gives 2 extra BR successes every round. The French purchase has shrunk to 0.3 K.

DISCUSS WITH YOUR TEAM

Why does the same school pay off in one country and not in another?

If your talent drains away, what else can your country do?

APPENDIX A

Quick Reference Card

Keep this card open during play.

Rule	Number
Budget per round	2 × W dice
Time	Round 1: 20 minutes. Later rounds: 10 minutes each.
BR die succeeds on	4–6
BR talent bonus	K ÷ 10, rounded down (only if you buy BR)
BR from other teams	0.75 × their totals
BR effective to unlock M / H	8 / 20
AR max dice per round	3 L, 2 M, 1 H
AR die cost	1, 1, 2, 2, 3, 3
AR level locks at	L 10 TP, M 20 TP, H 40 TP
Spies per round	Up to 2, 1 die each
Steal chance	Open target 1 in 3. Closed target never.
Saving	2/3 of unspent dice adds to W next round
Score weights	L × 1, M × 2, H × 3, then × TK multiplier, + 2√W

APPENDIX B

Questions for Your Team

The rulebook does not give a winning plan. Talk about these questions before round 1.

1. Should you pay for your own basic research, or use the research of other teams?
2. Open borders bring talent but invite theft. Which matters more for your country?
3. How much should you save when you do not know which round is the last?
4. When should you start H research? It pays most when your TK is 40 or more.
5. Which rivals can you rob? Which can you only watch?
6. What will rivals think about you? What will they do about it?

APPENDIX C

Team Roles

Each team member holds one portfolio. Each portfolio watches different numbers. Use this guide to know what to track and where to read about it.

Some numbers are an **output** for one role and an **input** for another. For example, Education produces talent (K), and talent turns into knowledge (TK) through basic research. Tech Strategy then uses TK as an input: high TK makes M and H research pay more.

Overview

Portfolio	Decides	Output to grow	Inputs to watch	Pages
Education	SE and TE purchases	K, and the TK that K produces	Talent flow, SE cost, talent limits (K 7 and 12)	5, 7, 9, 12, 15
Immigration	Open or closed borders (IM)	Talent flow number	Theft risk, TE bonus, rivals' W	11, 12
Tech Strategy	BR and AR dice	TP, and TK	TK, K, BR effective, AR cost, locks	8, 9, 10
National Security	Spy targets (SP)	Stolen TP and information on rivals	Rivals' IM, public log, rivals' BR	6, 11
Executive	Budget split, saving, final vote	Score	W, saving, score formula, time	6, 7, 13, 14

Who makes and who uses each number

Number	Made by	Used by	Pages
W	Executive (by saving)	Everyone (budget). Immigration (W pulls talent).	7, 12, 13
K	Education. Immigration (talent flow).	Tech Strategy (BR bonus, talent limits)	4, 9, 12, 15
TK	Tech Strategy (each BR success and AR die). Education (through the BR bonus).	Tech Strategy (bigger M and H rewards). Executive (score multiplier).	9, 10, 14
BR effective	Tech Strategy, and all rival teams	Tech Strategy (unlocks M at 8, H at 20)	9
TP	Tech Strategy (AR). National Security (steals).	Executive (score)	9, 11, 14
IM	Immigration	Education (TE bonus). National Security (theft risk).	11, 12

Education

- **Your job:** grow K. K gives extra BR successes (K ÷ 10), and each success adds TK.
- **Watch:** your K after each round, and your talent flow number. A number below 1 shrinks K every round.
- **Ask each round:** is K below 7 (M penalty) or 12 (H penalty)? Is SE or TE the better buy for us?

Immigration

- **Your job:** set the border policy. It changes talent flow and theft risk.
- **Watch:** your talent flow number, and which rivals spy on you (public log).
- **Ask each round:** is the talent we gain worth the theft risk? Open borders double the TE bonus.

Tech Strategy

- **Your job:** turn dice into TP. You choose BR and AR.
- **Watch:** TK (bigger M and H rewards), BR effective (unlocks), AR cost, and which levels are near their lock.
- **Ask each round:** can we reach M or H yet? Do we buy BR ourselves, or wait for rivals' BR?

National Security

- **Your job:** learn what rivals have, and steal TP when you can.
- **Watch:** which rivals are open (only they can be robbed), the public log, and rivals' BR totals.
- **Ask each round:** which rival and which level are worth a spy? Is our own country exposed?

Executive

- **Your job:** make the final call. Split the budget, decide how much to save, and submit on time.
- **Watch:** W, the score formula, and the clock. Savings in the last round are lost, and the last round is secret.
- **Ask each round:** does this plan grow our score, or only one portfolio's number?